Topic: For Loop in Python explaination:

A **for loop** in Python is used to **repeat a block of code for each item in a sequence**. A sequence can be a list, string, tuple, or a range of numbers.

Instead of writing the same code again and again, a for loop lets you do it automatically.

Basic Idea

Think of a for loop like this:

"Do this task for every item in the collection."

For example:

- Print numbers from 1 to 5
- Go through each character in a word
- Process each item in a list

Syntax of For Loop

```
for variable in sequence:
    # code to execute
```

Explanation:

- variable → stores each item one by one
- sequence → collection of values (like range, list, string)
- : → starts the loop block
- Indentation → code inside loop

Example : Using range()

```
for i in range(5):
    print(i)
```

Output:

```
0
1
2
3
4
```

Explanation:

- range(5) generates numbers from 0 to 4
- i takes each value one by one
- Loop runs 5 times

Why For Loop is Important

✔Reduces code repetition

✔Easy to work with collections

✔Very important in interviews

✔Used in real-world programming (data processing, AI, websites